IGOR RIVIN

Mathematician & Universalist

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in https://www.linkedin.com/in/igor-rivin-b95257249/



EXPERIENCE

Manager, Advanced Mathematics Group
Turing LLC
July 2026-present Princeton, NJ

Consultant
Turing LLC
July 2025-present Princeton, NJ

Generating challenging questions for LLM training.

Senior Advisor
Wolfram Inc
May 2023-present Princeton, NJ

Organized the 2023 ChainScience Conferences (the in-person and the virtual conferences).

Research Scientist
Edgestream LP
January 2020 – April 2023 Princeton, NJ

Working on understanding market data and developing quantitative strategies. Have worked on:

- Portfolio-based strategies.
- Optimal hedging strategies
- Machine learning based strategies
- Signal processing based strategies
- Software architecture
- Alternative data
- Extracting data from derivatives

Professor of Mathematics
Temple University
June 1999 – ongoing Philadelphia, PA

Director
The Cryptos Fund
October 2017 – October 2019 Zug, Switzerland

- Developed the `CCI30` cryptocurrency index - see `https://cci30.com`

Chief Strategy Officer
Accern, Inc
Jan 2017 – Jan 2018 New York, NY

- Developed trading strategies based on Accern's sentiment data.
- Developed the Natural Language parser.

LIFE PHILOSOPHY

"Fluctuat nec Mergitur."

STRENGTHS

Problem Solver | Curious | Generalist

Geometry | Number Theory | Probability | Computation | Statistics | Mathematica | Data Science | Quantitative Finance

LANGUAGES

English
Russian
French
Spanish
Portuguese

EDUCATION

Ph.D. in Mathematics
Princeton University
Sept 1981 – June 1986

- Advised by William P. Thurston
- Dissertation: "Geometry of convex polyhedra in hyperbolic 3-space."

B.Sc. (Hon) in Mathematics
University of Toronto
Sept 1977 – Sept 1981

ADVISORY BOARDS

Advisory board member
iVate.com
May 2019-present

Technical Advisory Board member
Accern.com
January 2015-present

Research Director

Ticklr, Inc

May 2017 – May 2019 London, England

- Developed Blockchain technologies for Social Media.

Regius Professor of Mathematics

Univesity of St Andrews

May 2015 – May 2017 St Andrews, Scotland

Principal

Meteque Holdings

2006-2016 Princeton, New Jersey

- Built a hedge fund (Samsara Investment Partners) which outperformed the market by wide margin for eight years, despite being a one man operation.

Member, School of Mathematics

Institute for Advanced Study

September 2010-July 2011 Princeton, New Jersey

Berlin Mathematics Graduate School Professor

September 2011-Dec 2011

Berlin, Germany

Visiting Professor

Stanford University

2007 Stanford, California

ICERM professor of Mathematics

Brown University

2013-2014 Providence, Rhode Island

Lady Davis Professor

Hebrew University

2006 Jerusalem, Israel

Consultant

Susquehanna International Group

2005-2006 Bala Cynwyd, Pennsylvania

- Worked in the statistical arbitrage group; developed a number of successful medium-hold-time trading strategies.

Consultant

Equities Trading Laboratory, Morgan Stanley

2004 New York, New York

- Developed a new market impact model.
- Developed a new options trading strategy.

EPSRC Advanced Fellow (with rank of Reader)

University of Manchester

1999-2002 Manchester, UK

Warwick Research Fellow

University of Warwick

1995-1998 Coventry, UK

Olga Tausski-John Todd Instructor

Caltech

1995-1998 Pasadena, California

Research Fellow

University of Melbourne

1994-1995 Melbourne, Australia

Member, School of Mathematics

Institute for Advanced Study

1993-1994 Princeton, New Jersey

Consultant

NEC Research Institute

1991-1993 Princeton, New Jersey

- Jointly with Mike Treacy developed the subject of *hypothetical zeolites*
- Jointly with Srimat Chakradhar developed new methodology for VLSI testing.
- Made breakthroughs in computational and discrete geometry.
- Made significant contributions to Machine Learning (jointly with Yuh-Dauh Lyuuu.)

Director of Advanced Development

Wolfram Research Inc

Jan 1989-March 1991 Champaign, Illinois

- Developed large parts of the *Mathematica* kernel, including 3-D graphics, linear algebra, power series, limits, big number multiplication, solution of algebraic equations, and others.
- Supervised a group of six developers (all around the world).

Director of Applications Development, the QLISP project

Stanford University Computer Science Department

Jan 1987-Dec 1988 Stanford, California

- worked on the QLISP system for parallel symbolic computation (with John McCarthy).
- Developed algorithms for symbolic computation and task scheduling.

- Developed new methods for constraint propagation (with Ramin Zabih).

Visiting Professor

Institut des Hautes Études Scientifiques

1985-1986 Paris, France

PUBLICATIONS

Journal Articles

- Kolpakov, Alexander and Igor Rivin. "DiRe - JAX: A JAX based Dimensionality Reduction Algorithm for Large-scale Data". In: *The Journal of Open Source Software* (2025). DOI: 10.21105/joss.08264. arXiv: 2503.03156.
- – "Graphem - JAX: Node Influence Maximization via Geometric Embeddings". In: *The Journal of Open Source Software* (2025). DOI: 10.21105/joss.08855.
- Rivin, Igor. "Computing Periodic Billiard Orbits in L^p Balls via Newton's Method and Smale's alpha-Criterion". In: *arXiv preprint arXiv:2507.22251* (2025).
- – "Random Graphs from Random Matrices". In: *Journal of Experimental Mathematics* (2025). DOI: 10.56994/jxm.001.002.009. arXiv: 1910.08253.
- – "(More) Robust Variance Estimation". In: *Available at SSRN 4767015* (2024).
- Rivin, Igor and Naser T Sardari. "Quantum Chaos on Random Cayley Graphs of SL 2 [Z/p Z]". in: *Experimental Mathematics* 28.3 (2019), pp. 328–341.
- – "Quantum Chaos on Random Cayley Graphs of SL 2[Z/pZ]". in: *Experimental Mathematics* (2017). DOI: 10.1080/10586458.2017.1403982.
- Fuchs, Elena and Igor Rivin. "Generic thinness in finitely generated subgroups of SL(n, Z)". in: *arXiv preprint arXiv:1506.01735* (2016), rnw136.
- – "Generic Thinness in Finitely Generated Subgroups of $\mathrm{SL}_n(\mathbb{Z})$". In: *International Mathematics Research Notices* (2016). DOI: 10.1093/imrn/rnw136.
- Rivin, Igor. "Spectral Experiments+". In: *Experimental Mathematics* (2016). DOI: 10.1080/10586458.2015.1060908. arXiv: 1410.6771.
- Pemantle, Robin, Yuval Peres, and Igor Rivin. "Four random permutations conjugated by an adversary generate Sn with high probability". In: *Random Structures and Algorithms* (2015). DOI: 10.1002/rsa.20632. arXiv: 1412.3781.
- Rivin, Igor. "How to pick a random integer matrix? (and other questions)". In: *Mathematics of Computation* (2015). DOI: 10.1090/mcom/2986. arXiv: 1312.4607.
- – "Generic phenomena in groups: some answers and many questions". In: *Thin Groups and Superstrong Approximation* (2014). DOI: 10.1017/9781009701228.015. arXiv: 1211.6509.
- – "Some Thoughts on the Teaching of Mathematics–Ten Years Later". In: *Notices of the American Mathematical Society* (2014). DOI: 10.1090/noti1131. arXiv: 1401.0828.
- Malestein, Justin, Igor Rivin, and Louis Theran. "Topological designs". In: *Geometriae Dedicata* (2013). DOI: 10.1007/s10711-012-9827-9. arXiv: 1008.3710.
- Treacy, M.M.J. et al. "Flexibility mechanisms in ideal zeolite frameworks". In: *Philosophical Transactions of the Royal Society A Mathematical Physical and Engineering Sciences* (2013). DOI: 10.1098/rsta.2012.0036.
- Rivin, Igor. "Geodesics with one self-intersection, and other stories". In: *Advances in Mathematics* (2012). DOI: 10.1016/j.aim.2012.07.018. arXiv: 0901.2543.
- Kapko, Vitaliy et al. "Density of Mechanisms within the Flexibility

HONORS AND AWARDS

Top Five by Reputation
MathOverflow
2016–ongoing · Worldwide

Fellow
American Mathematical Society
2015 · USA

ICERM Professor
Brown University
2013-2014 · Providence, RI

BMS Professor
Berlin Mathematical School
2011 · Berlin, Germany

Lady Davis Fellow
Hebrew University
2006 · Jerusalem, Israel

Advanced Research Fellow
EPSRC
1999 · UK

Whitehead Prize
London Mathematical Society
1998 · UK

Plenary Address
Nordic Mathematics Congress
1991 · Sweden

First Prize
Canadian Mathematical Olympiad
1977 · Canada

EDITORIAL BOARDS

International Mathematics Research Notices
2015-present

Geometriae Dedicata
2004-2018

New York Journal of Mathematics
2016-2018

Experimental Mathematics

Window of Zeolites". In: *Physical Review Letters* (2011). DOI: 10.1103/physrevlett.107.164304.

- Droste, Manfred and Igor Rivin. "On extension of coverings". In: *Bulletin of the London Mathematical Society* (2010). DOI: 10.1112/blms/bdq068. arXiv: 0901.3594.
- Rivin, Igor. "Growth in free groups (and other stories)—twelve years later". In: *Illinois Journal of Mathematics* (2010). DOI: 10.1215/ijm/1299679752.
- – "Zariski Density and Genericity". In: *International Mathematics Research Notices* (2010). DOI: 10.1093/imrn/rnq043.
- Treacy, M.M.J., Martin D. Foster, and Igor Rivin. "ChemInform Abstract: Towards a Catalogue of Designer Zeolites". In: *ChemInform* (2010). DOI: 10.1002/chin.201017218.
- Mohar, Bojan and Igor Rivin. "Simplices and Spectra of Graphs". In: *Discrete & Computational Geometry* (2009). DOI: 10.1007/s00454-009-9206-y. arXiv: 0901.3284.
- Rivin, Igor. "Walks on graphs and lattices – effective bounds and applications". In: *Forum Mathematicum* (2009). DOI: 10.1515/forum.2009.034. arXiv: math/0703533.
- Kapovich, Ilya and Igor Rivin. "On the absence of McShane-type identities for the outer space". In: *Journal of Algebra* (2008). DOI: 10.1016/j.jalgebra.2008.08.024. arXiv: math/0602291.
- Rivin, Igor. "Asymptotics of convex sets in Euclidean and hyperbolic spaces". In: *Advances in Mathematics* (2008). DOI: 10.1016/j.aim.2008.11.014. arXiv: 0801.0077.
- – "Walks on groups, counting reducible matrices, polynomials, and surface and free group automorphisms". In: *Duke Mathematical Journal* (2008). DOI: 10.1215/00127094-2008-009. arXiv: math/0703532.
- Kapovich, Ilya et al. "Densities in free groups and $\mathbb{Z}^k$, Visible Points and Test Elements". In: *Mathematical Research Letters* (2007). DOI: 10.4310/mrl.2007.v14.n2.a9. arXiv: math/0507573.
- Rivin, Igor. "Surface area and other measures of ellipsoids". In: *Advances in Applied Mathematics* (2007). DOI: 10.1016/j.aam.2006.08.009. arXiv: math/0403375.
- – "Volumes of degenerating polyhedra — on a conjecture of J. W. Milnor". In: *Geometriae Dedicata* (2007). DOI: 10.1007/s10711-007-9217-x. arXiv: math/0512065.
- – "A lesson from virtual zeolites". In: *Nature Materials* (2006). DOI: 10.1038/nmat1792.
- – "Geometric simulations: a lesson from virtual zeolites". In: *Nature materials* 5.12 (2006), pp. 931–932.
- Foster, Martin D. et al. "A geometric solution to the largest-free-sphere problem in zeolite frameworks". In: *Microporous and Mesoporous Materials* (2005). DOI: 10.1016/j.micromeso.2005.08.025.
- Foster, Martin D. et al. "A systematic topological search for the framework of ZSM-10". In: *Journal of Applied Crystallography* (2005). DOI: 10.1107/s0021889805026038.
- Kapovich, Ilya et al. "Asymptotic density in free groups and Z k, visible points and test elements". In: *arXiv preprint math.GR/0507573* (2005).
- Rivin, Igor. "A Simpler Proof of Mirzakhani's Simple Curve Asymptotics". In: *Geometriae Dedicata* (2005). DOI: 10.1007/s10711-005-7153-1. arXiv: math/0512066.
- – "On Some Mean Matrix Inequalites of Dynamical Interest". In: *Communications in Mathematical Physics* (2005). DOI: 10.1007/s00220-004-1282-5. arXiv: math/0312048.
- – "A Remark on 'Counting Primitive Elements in Free Groups' (By J. Burillo and E. Ventura)". In: *Geometriae Dedicata* (2004). DOI:

10.1023/b:geom.0000049102.82732.1f. arXiv: math/0302083.

- Rivin, Igor. “Symmetrized Chebyshev polynomials”. In: *Proceedings of the American Mathematical Society* (2004). DOI: 10.1090/s0002-9939-04-07664-6. arXiv: math/0301241.
- Treacy, M.M.J. et al. “Enumeration of periodic tetrahedral frameworks. II. Polynodal graphs”. In: *Microporous and Mesoporous Materials* (2004). DOI: 10.1016/j.micromeso.2004.06.013.
- Komarova, Natalia L. and Igor Rivin. “Harmonic mean, random polynomials and stochastic matrices”. In: *Advances in Applied Mathematics* (2003). DOI: 10.1016/s0196-8858(03)00023-x. arXiv: math/0105236.
- Rivin, Igor. “Combinatorial optimization in geometry”. In: *Advances in Applied Mathematics* (2003). DOI: 10.1016/s0196-8858(03)00093-9. arXiv: math/9907032.
- Jakobson, Dmitry and Igor Rivin. “Extremal metrics on graphs I”. in: *Forum Mathematicum* (2002). DOI: 10.1515/form.2002.002. arXiv: math/0001169.
- Rivin, Igor. “Counting cycles and finite dimensional L norms”. In: *Advances in Applied Mathematics* (2002). DOI: 10.1016/s0196-8858(02)00037-4.
- – “Simple Curves on Surfaces”. In: *Geometriae Dedicata* (2001). DOI: 10.1023/a:1012010721583. arXiv: math/9907041.
- Rivin, Igor and Jean-Marc Schlenker. “The Schläfli formula in Einstein manifolds with boundary”. In: *Electronic Research Announcements of the American Mathematical Society* (1999). DOI: 10.1090/s1079-6762-99-00057-8.
- Rivin, Igor and John H. Lindsey. “A Similarity Criterion: 10462”. In: *American Mathematical Monthly* (1998). DOI: 10.2307/2589266.
- Cooper, Daryl and Igor Rivin. “Combinatorial scalar curvature and rigidity of ball packings”. In: *Mathematical Research Letters* (1996). DOI: 10.4310/mrl.1996.v3.n1.a5.
- Rivin, Igor. “A Characterization of Ideal Polyhedra in Hyperbolic 3-Space”. In: *Annals of Mathematics* (1996). DOI: 10.2307/2118652.
- Beckwith, David et al. “Problems: 10459-10465”. In: *American Mathematical Monthly* (1995). DOI: 10.2307/2974774.
- McShane, Greg and Igor Rivin. “Geometry of geodesics and a norm on homology”. In: *International Mathematics Research Notices* 2 (1995), pp. 61–69.
- Rivin, Igor. “Euclidean Structures on Simplicial Surfaces and Hyperbolic Volume”. In: *Annals of Mathematics* (1994). DOI: 10.2307/2118572.
- Rivin, Igor, Ilan Vardi, and Paul Zimmermann. “The n -Queens Problem”. In: *American Mathematical Monthly* (1994). DOI: 10.1080/00029890.1994.11997004.
- Hodgson, Craig D. and Igor Rivin. “A characterization of compact convex polyhedra in hyperbolic 3-space”. In: *Inventiones mathematicae* (1993). DOI: 10.1007/bf01231281.
- Rivin, Igor. “On geometry of convex ideal polyhedra in hyperbolic 3-space”. In: *Topology* (1993). DOI: 10.1016/0040-9383(93)90039-x.
- Hodgson, Craig D., Igor Rivin, and Warren D. Smith. “A characterization of convex hyperbolic polyhedra and of convex polyhedra inscribed in the sphere”. In: *Bulletin of the American Mathematical Society* (1992). DOI: 10.1090/s0273-0979-1992-00303-8. arXiv: math/9210218.
- Lyuu, Yuh-Dauh and Igor Rivin. “Tight Bounds on Transition to Perfect Generalization in Perceptrons”. In: *Neural Computation* (1992). DOI: 10.1162/neco.1992.4.6.854.
- Rivin, Igor and Ramin Zabih. “A dynamic programming solution to the n-queens problem”. In: *Information Processing Letters* (1992). DOI: 10.1016/0020-0190(92)90168-u.

- Grinberg, Eric L. and Igor Rivin. “Infinitesimal Aspects of the Busemann-Petty Problem”. In: *Bulletin of the London Mathematical Society* (1990). DOI: 10.1112/blms/22.5.478.
- Rivin, Igor and Ramin Zabih. “An algebraic approach to constraint satisfaction problems”. In: (1989).

Books

- Bassino, Frédérique et al. *Complexity and Randomness in Group Theory*. 2020. DOI: 10.1515/9783110667028.
- Rivin, Igor, C. P. Rourke, and Caroline Series. *The Epstein birthday schrift : papers dedicated to David Epstein on the occasion of his 60th birthday, 16 May 1997*. Japan Society of Medical Entomology and Zoology, 1998.

Book Chapters

- Jakobson, Dmitry et al. “Rank and Bollobás-Riordan polynomials: Coefficient measures and zeros”. In: *Contemporary mathematics - American Mathematical Society*. American Mathematical Society, 2019. DOI: 10.1090/conm/739/14894.
- Pemantle, Robin and Igor Rivin. “The Distribution of Zeros of the Derivative of a Random Polynomial”. In: *Advances in Combinatorics*. 2013. DOI: 10.1007/978-3-642-30979-3_14. arXiv: 1109.5975.
- Rivin, Igor. “The Moment Zeta Function and Applications”. In: *Developments in mathematics*. Springer Nature, 2012. DOI: 10.1007/978-1-4614-4075-8_22. arXiv: math/0201109.
- Treacy, M.M.J., Martin D. Foster, and Igor Rivin. “Towards a Catalogue of Designer Zeolites”. In: *Royal Society of Chemistry eBooks*. Royal Society of Chemistry, 2007. DOI: 10.1039/9781847558183-00208.
- Jakobson, Dmitry et al. “Eigenvalue Spacings for Regular Graphs”. In: *The IMA volumes in mathematics and its applications*. Institute of Electrical and Electronics Engineers, 1999. DOI: 10.1007/978-1-4612-1544-8_12. arXiv: hep-th/0310002.

Conference Proceedings

- Kolpakov, Alexander and Igor Rivin. “Discovering temporal and personality aspects in meager and highly variable text samples”. In: *Proceedings of the 2024 8th International Conference on Natural Language Processing and Information Retrieval*. 2024. DOI: 10.1145/3711542.3711574.
- Curto, Carina, Joshua Paik, and Igor Rivin. “Betti Curves of Rank One Symmetric Matrices”. In: *Lecture notes in computer science*. 2021. DOI: 10.1007/978-3-030-80209-7_69. arXiv: 2103.00761.
- Alam, Ashraful, Igor Rivin, and Ileana Streinu. “Outerplanar Graphs and Delaunay Triangulations”. In: *Lecture notes in computer science*. 2012. DOI: 10.1007/978-3-642-27654-5_25.
- Rivin, Igor. “Some properties of the conjugacy class growth function”. In: *Contemporary mathematics - American Mathematical Society*. 2004. DOI: 10.1090/conm/360/06573.
- Rivin, Igor and Srimat Chakradhar. “Discrete test generation by continuous methods”. In: *Proceedings of IEEE VLSI Test Symposium*. 2002. DOI: 10.1109/vtest.1994.292327.
- Almgren, Frederic J. and Igor Rivin. “The mean curvature integral is invariant under bending”. In: *Geometry and topology monographs*. 1998. DOI: 10.2140/gtm.1998.1.1. arXiv: math/9810183.

- Treacy, M.M.J., S. RAO, and Igor Rivin. “A COMBINATORIAL METHOD FOR GENERATING NEW ZEOLITE FRAMEWORKS”. in: *Elsevier eBooks*. 1993. DOI: 10.1016/b978-1-4832-8383-8.50044-1.

Technical Reports

- Rabinov, Arkady and Igor Rivin. *Programming in Qlisp: A Case Study*. Tech. rep. DTIC Document, 1989.
- Rivin, Igor. *Customizing the Definite Integrator, Part II*. tech. rep. 1987.
- – *Customizing the Definite Integrator, Part I*. tech. rep. 1986.

Journal Blog Posts

- Rivin, Igor. *A Conversation with Grok 3 on US Politics and WW3*. https://impurethoughts.substack.com/p/a-conversation-with-grok-3-on-us. Accessed: 2025-02-28. Feb. 2025.
- – *A Quick Look at Image-to-Video*. https://impurethoughts.substack.com/p/a-quick-look-at-image-to-video. Accessed: 2025-02-28. Feb. 2025.
- – *And Now for Graph Embedding*. https://impurethoughts.substack.com/p/and-now-for-graph-embedding. Accessed: 2025-02-28. Feb. 2025.
- – *From Surfaces to Curves*. https://impurethoughts.substack.com/p/from-surfaces-to-curves. Accessed: 2025-02-28. Feb. 2025.
- – *Graphs and Sampling, by Way of Parallelism*. https://impurethoughts.substack.com/p/graphs-and-sampling-by-way-of-parallelism. Accessed: 2025-02-28. Feb. 2025.
- – *How to Draw a Surface*. https://impurethoughts.substack.com/p/how-to-draw-a-surface. Accessed: 2025-02-28. Feb. 2025.
- – *Spikey: A Million Years Later*. https://impurethoughts.substack.com/p/spikey-a-million-years-later. Accessed: 2025-02-28. Feb. 2025.
- – *The Chaos Game in 2025*. https://impurethoughts.substack.com/p/the-chaos-game-in-2025. Accessed: 2025-02-28. Feb. 2025.
- – *Tubes*. https://impurethoughts.substack.com/p/tubes. Accessed: 2025-02-28. Feb. 2025.

Preprints

- Rivin, Igor and Carlo Scevola. *An Investable Cryptocurrency Index*. eprint: 3154706.
- Kolpakov, Alexander and Igor Rivin. *DiRe-RAPIDS: Topology-faithful dimensionality reduction at scale*. 2026. arXiv: 2604.25209.
- – *Discriminant Varieties for Stick Knots and Links*. 2026. arXiv: 2608.01277.
- Rivin, Igor. *Linear recurrences and rational Lambert series*. 2026. arXiv: 2604.25151.
- – *Monte-Carlo Irreducibility and Imprimitivity Detection of Polynomials over $\mathbb{Q}$*. 2026. arXiv: 2602.00944.
- – *Permanents of matrix ensembles: computation, distribution, and geometry*. 2026. arXiv: 2602.10141.

- Rivin, Igor. *Probing Structural Mathematical Reasoning in Language Models with Algebraic Trapdoors*. 2026. arXiv: 2605.04352.
- Kolpakov, Alexander and Igor Rivin. *Fast Geometric Embedding for Node Influence Maximization*. 2025. arXiv: 2506.07435.
- Rivin, Igor. *Computational Resolution of Hadamard Product Factorization for 4×4 Matrices*. 2025. arXiv: 2508.14901.
- – *Computing Periodic Billiard Orbits in L^p Balls via Newton's Method and Smale's -Criterion*. 2025. arXiv: 2507.22251.
- – *Counting Matrices in SL3(Z) with Fixed Completely Split Character Polynomial: Preliminary Upper Bounds*. 2025. arXiv: 2507.15129.
- – *From Euler to Today: Universal Mathematical Fallibility A Large-Scale Computational Analysis of Errors in ArXiv Papers*. 2025. arXiv: 2511.10543.
- – *Maximal Volume Ideal Polyhedra and the Arithmetic Angle Phenomenon*. 2025. arXiv: 2512.10087.
- Kolpakov, Alexander and Igor Rivin. *A ripple in time: a discontinuity in American history*. 2023. arXiv: 2312.01185.
- Rivin, Igor. *A Currency Addendum to The Volatility Smile Quantified, and Other Stories'*. 2023. SSRN: 4519663.
- – *The Volatility Smile Quantified and Other Stories*. 2023. SSRN: 4519663.
- – *A Bayesian addendum to the Bibliometric Analysis of Senior US Mathematics Faculty*. 2021.
- Paik, Joshua and Igor Rivin. *Bibliometric Analysis of Senior US Mathematics Faculty*. 2020. arXiv: 2008.11196.
- – *Data Analysis of the Responses to Professor Abigail Thompson's Statement on Mandatory Diversity Statements*. 2020. arXiv: 2001.00670.
- Rivin, Igor. *Adventures in Financial Time Series*. 2019. SSRN: 3344910.
- – *Experiments with the Census*. 2019. arXiv: 1903.09532.
- – *Fear Universality and Doubt in Asset price movements*. 2018. arXiv: 1803.07138.
- – *The end of the West - a Prologue*. 2018. SSRN: 3185162.
- – *What is the Sharpe Ratio, and how can everyone get it wrong?* 2018. arXiv: 1802.04413.
- Rivin, Igor and Carlo Scevola. *The CCI30 Index*. 2018. arXiv: 1804.06711.
- Rivin, Igor and Andrew Seaton. *An analysis of a war*. 2018. SSRN: 3150129.
- Rivin, Igor and Naser T. Sardari. *Quantum Chaos on random Cayley graphs of $\mathrm{SL}_2[\mathbb{Z}/p\mathbb{Z}]$*. 2017. arXiv: 1705.02993.
- Rivin, Igor. *Random space and plane curves*. 2016. arXiv: 1607.05239.
- Fuchs, Elena and Igor Rivin. *Generic thinness in finitely generated subgroups of $SL_n(\mathbb{Z})$*. 2015. arXiv: 1506.01735.
- Rivin, Igor. *Galois Groups of Generic Polynomials*. 2015. arXiv: 1511.06446.
- – *Some experiments on Bateman-Horn*. 2015. arXiv: 1508.07821.
- – *On Basmajian's identities, and other stories*. 2014. arXiv: 1404.1583.
- – *Statistics of Random 3-Manifolds occasionally fibering over the circle*. 2014. arXiv: 1401.5736.
- – *Large Galois groups with applications to Zariski density*. 2013. arXiv: 1312.3009.
- – *Rigidity of Fibering*. 2011. arXiv: 1106.4595.
- – *Walks on Free Groups and other Stories – twelve years later*. 2011. arXiv: 1106.5947.
- – *Golden-Thompson from Davis*. 2010. arXiv: 1010.2193.
- – *Counting Reducible Matrices, Polynomials, and Surface and Free Group Automorphisms*. 2006. arXiv: math/0604489.

- Rivin, Igor. *Continuity of volumes – on a generalization of a conjecture of J. W. Milnor*. 2005. arXiv: math/0502543.
- – *Extra-large metrics*. 2005. arXiv: math/0509320.
- – *Triangulations into Groups*. 2005. arXiv: math/0510613.
- – *Estimates and identities for the average distortion of a linear transformation*. 2004. arXiv: math/0412260.
- Cejtin, Henry and Igor Rivin. *A property of alternating groups*. 2003. arXiv: math/0303036.
- Grabovsky, Yury, Omar Hijab, and Igor Rivin. *Differentiability of functions of matrices*. 2003. arXiv: math/0310086.
- Rivin, Igor. *An extended correction to "Combinatorial Scalar Curvature and Rigidity of Ball Packings," (by D. Cooper and I. Rivin)*. 2003. arXiv: math/0302069.
- – *An inequality on Chebyshev polynomials*. 2003. arXiv: math/0301210.
- – *Simple estimates for ellipsoid measures*. 2003. arXiv: math/0306085.
- – *Some observations on the simplex*. 2003. arXiv: math/0308239.
- – *Spheres and Minima*. 2003. arXiv: math/0305252.
- – *Surface Area of Ellipsoids*. 2003. arXiv: math/0306387.
- – *A multidimensional Law of Sines*. 2002. arXiv: math/0211261.
- – *Another Simple Proof of a Theorem of Chandler Davis*. 2002. arXiv: math/0208223.
- – *The amplitude modulation transform*. 2002. arXiv: math/0212199.
- – *The performance of the batch learner algorithm*. 2002. arXiv: cs/0201009.
- Komarova, Natalia and Igor Rivin. *Mathematics of learning*. 2001. arXiv: math/0105235.
- Rivin, Igor. *Counting cycles and finite dimensional Lp norms*. 2001. arXiv: math/0111106.
- – *Yet another zeta function and learning*. 2001. arXiv: cs/0107033.
- McShane, Greg and Igor Rivin. *A norm on homology of surfaces and counting simple geodesics*. 2000. arXiv: math/0005222.
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